

R2 multi-scale bookkeeping: the power-sum lemma closes the N-counting; the D-loss reduces to one decoupling invocation

TECT verification-first repository · claim B5-BEYOND-LAYER-BOUND

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1. Purpose and scope

The arbitrary- Q DR-2 conditional chain (dr2-cross-scale-induction v1.1) rests on two cited-not-reproduced residuals: R1 (local-to-global), now a written proof (R-034, dr2-t030-frontier-consolidation v1.1), and R2 (multi-scale bookkeeping). This note attacks R2. **Scope:** the bookkeeping of the one-step inequality

$$E_+(Q) \leq D^2 \left(\sum_{\theta} E_+(Q_{\theta})^{1/2} \right)^2$$

iterated across scales, where the $\delta^{1/2}$ -caps θ partition Q , D is the ℓ^2 -decoupling constant at one scale, and (R-023, affine invariance) each Q_{θ} rescales loss-free to a unit paraboloid patch. **Not claimed:** an unconditional $N^{2+\varepsilon}$; any flip.

2. Pre-registered gates

G1: if the per-generation losses multiply unboundedly and no summable- ε schedule or one-shot replacement removes the geometry dependence, record an HONEST PARTIAL with the residual named precisely. G2: success = a complete proof, every constant chase explicit, graded T6 PROVED CONDITIONAL on Bourgain–Demeter ONLY. **Outcome: G1.**

3. The N-counting is loss-free (power-sum lemma, PROVED)

Lemma 1 (Power-sum / superadditivity). *For $p \geq 1$ and nonnegative integers N_{θ} with $\sum_{\theta} N_{\theta} = N$,*

$$\sum_{\theta} N_{\theta}^p \leq \left(\sum_{\theta} N_{\theta} \right)^p = N^p,$$

with equality iff at most one N_{θ} is nonzero.

Proof. $x \mapsto x^p$ is superadditive on $[0, \infty)$ for $p \geq 1$: $(a + b)^p \geq a^p + b^p$ (convexity + $0^p = 0$); induct on the number of parts. Strictness unless one term carries all the mass. $\square$

Consequence. Substituting the induction hypothesis $E_+(Q_{\theta}) \leq C N_{\theta}^{2+\varepsilon}$ into the one-step inequality,

$$E_+(Q) \leq D^2 \left(\sum_{\theta} (C N_{\theta}^{2+\varepsilon})^{1/2} \right)^2 = D^2 C \left(\sum_{\theta} N_{\theta}^{1+\varepsilon/2} \right)^2 \leq D^2 C (N^{1+\varepsilon/2})^2 = D^2 C N^{2+\varepsilon},$$

the last step by the Lemma with $p = 1 + \varepsilon/2 \geq 1$. The N -dependence is therefore carried with NO loss at ANY split, balanced or not – the maximally-unbalanced partition $(N - 1, 1)$ obeys the Lemma exactly (machine: assert 1, 400 random partitions plus $(N - 1, 1)$ for $N \leq 499$, worst lhs/rhs ≤ 1). This DISSOLVES the unbalanced-tree fear for the N -counting: the only thing that can blow up is the D -factor.

4. The D-counting: naive tree fails; submultiplicativity repairs

The obstruction (naive fixed-scale tree). Recursing the one-step inequality at a FIXED scale δ_0 multiplies a fixed D_0 per generation. The maximally-unbalanced tree (peel one point per split) has depth $G = N - 1$, so the naive bound carries $D_0^{2G} = D_0^{2(N-1)}$ – EXPONENTIAL in N , not $N^{O(\varepsilon)}$ (machine: assert 3, $N = 30$ gives $D_0^{2 \cdot 29} \approx 2.9 \times 10^{17} = e^{40.2}$). Any R2 proof that silently assumes balanced splits or depth $O(\log N)$ is therefore WRONG; this is the documented failure mode.

The repair (submultiplicativity / one-shot). The ℓ^2 -decoupling constant is submultiplicative across scales, $D(\delta_1 \delta_2) \leq C_{\text{sub}} D(\delta_1) D(\delta_2)$, and the Bourgain–Demeter theorem packages the entire scale-1-to- δ passage as a SINGLE application with a SINGLE loss $D(\delta) \leq_\varepsilon \delta^{-\varepsilon}$ (the multi-scale induction is INTERNAL to its proof). Decoupling directly from scale 1 to the finest relevant scale s (the min separation, where every cap holds ≤ 1 point = base case) gives

$$E_+(Q) \leq_\varepsilon s^{-2\varepsilon} N^2,$$

ONE loss, independent of tree shape. For polynomially-separated Q ($s \geq N^{-C}$) this is $s^{-2\varepsilon} \leq N^{2C\varepsilon} = N^{O(\varepsilon)}$ (machine: assert 4, $s = N^{-3}$ gives the fixed power $N^{6\varepsilon}$, log-dominating the exponential naive factor at $N = 1000$: 4.1 vs 1384.9).

5. The residual, named precisely (gate G1)

For ARBITRARY (not polynomially-separated) Q , the min separation s can be sub-polynomial, and then $s^{-2\varepsilon}$ is NOT $N^{O(\varepsilon)}$: the adversarial $s = e^{-N}$ gives one-shot exponent $2\varepsilon N / \log N$, which GROWS unboundedly ($1.8 \rightarrow 10.5 \rightarrow 74.9$ at $N = 30/300/3000$; machine: assert 5). So telescoping ALONE does not deliver the uniform-in-geometry bound. The single remaining ingredient is exactly:

[BD-DISCRETE]: the Bourgain–Demeter discrete well-separated reduction,

which pigeonholes the $O(\log N)$ relevant scales and reduces the arbitrary configuration to the separated base case with a single $N^{O(\varepsilon)}$ loss – equivalently, accepting polynomial separation $s \geq N^{-O(1)}$ after the affine normalization. This is standard decoupling machinery (the SAME single $\delta^{-\varepsilon}$ Bourgain–Demeter prove), invoked, not reproduced.

Net effect on R2. Before: R2 = “vague multi-scale bookkeeping over $O(\log N)$ levels.” After: the N -counting is a PROVED loss-free power-sum lemma; the D -counting is a single [BD-DISCRETE] invocation. R2 is reduced to ONE named, standard ingredient. With R1 proved, the arbitrary- Q chain rests on: the separated base case (T6 cond. on BD) + [BD-DISCRETE]. The chain’s honest grade is unchanged – **T6 PROVED CONDITIONAL on Bourgain–Demeter** – but its residual set is now a SINGLE clean citation.

6. Devil’s-advocate / self-adversarial review (CLAUDE.md 6.3)

- (α) “*The power-sum lemma is trivial – does it really carry the unbalanced-tree worry?*” VALID, and that is the point: the worry was MISPLACED on the N -counting (where it is dissolved by a one-line convexity fact) and BELONGS to the D -counting (assert 3), which submultiplicativity handles. Separating the two is the contribution; neither half is deep, but the conflation was what kept R2 vague.
- (β) “*Submultiplicativity of the decoupling constant for C^2 graph patches is itself cited.*” UPHELD as the residual: yes – that is precisely [BD-DISCRETE], named as the single remaining ingredient. The note does NOT claim to reprove Bourgain–Demeter; it claims to reduce R2 TO it, cleanly.

- (γ) “Affine rescaling to a unit paraboloid patch might lose the curvature that makes decoupling work.”
 DISMISSED: parabolic rescaling maps a paraboloid patch to a paraboloid patch EXACTLY (R-023; dr2-affine-invariance), so curvature is normalized, not lost – the flat-GAP escape (large planar additive energy) is illusory because the normal coordinates impose the curved constraint at every scale.

Quantitative sanity check (mandatory). The power-sum slack is exactly 0 at the trivial partition and strictly positive otherwise (assert 2: (100,100) slack 0.034); the naive vs one-shot comparison is in log scale to avoid the $D_0^{2(N-1)}$ overflow ($N=1000$: 1384.9 vs 4.1); the adversarial exponent growth is monotone and unbounded (assert 5). All five checks PASS, exit 0; the verdict string in the JSON artefact records G1-PARTIAL.

7. Result footer

Result ID:	B5-BEYOND-LAYER-BOUND / DR-2 R2 multi-scale bookkeeping (T-030 conditional chain)
Precise statement:	R2 PARTIAL ADVANCE (gate G1). N-counting: the power-sum lemma $\sum N_{\theta^p} \leq N^p$ ($p \geq 1$) is PROVED, so the recursion's N-dependence is loss-free at any split balance (the unbalanced-tree fear is dissolved). D-counting: a naive fixed-scale tree accumulates $D^{2(N-1)}$ (exponential, demonstrated), but submultiplicativity collapses it to one critical-scale loss $s^{-2\epsilon} N^2$. Residual NAMED: [BD-DISCRETE] (Bourgain-Demeter discrete well-separated reduction) removes the geometry-dependent $s^{-\epsilon}$; equivalently poly-separation $s \geq N^C$. R2 reduced from vague bookkeeping to ONE clean invocation.
Scope:	the multi-scale bookkeeping residual only; with R1 (R-034) proved, the arbitrary-Q chain rests on separated base + [BD-DISCRETE].
Evidence grade:	PARTIAL ADVANCE; the chain stays T6 PROVED CONDITIONAL on Bourgain-Demeter (residual set tightened {R1,R2} $\rightarrow$ single [BD-DISCRETE]).
Reproduction command:	python codes/vacuum/dr2_t030_r2_bookkeeping.py
Falsifier:	a partition with $\sum N_{\theta^p} > N^p$ ($p \geq 1$); a fixed-scale tree bound below D^{2G} on the unbalanced tree; a claim that telescoping alone gives $N^{2+\epsilon}$ for sub-polynomial s .
Tier before / after:	none -- operator decides. B5 stays T6-conditional; T-030 stays OPEN for unconditional $N^{2+\epsilon}$; arbitrary-Q chain stays T6 PROVED CONDITIONAL on BD.
No-overclaim statement:	R2 is REDUCED, not closed; [BD-DISCRETE] is cited not reproduced; no unconditional bound; no flip.
Next required action:	operator-stated T7 routes (one required): (i) adopt [BD-DISCRETE] as a standard black box $\rightarrow$ fix the arbitrary-Q chain as a conditional theorem (T6-BDconditional, the current grade), OR (ii) reprove the BD discrete reduction in-bundle $\rightarrow$ remove the last conditional residual of T-030. T-030 unconditional remains the open frontier.